

Experiment – 16
CSA1717 - Artificial Intelligence

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Aim:

To write a Python program to implement list methods such as add, append, extend, and delete.

Algorithm:

1. Start the program.
2. Create an initial list.
3. Add elements using append() and insert().
4. Extend the list using extend() to add multiple elements.
5. Delete elements using remove(), pop(), and del.
6. Display the list after each operation.
7. Stop the program.

Source code:

```
# Program to implement list methods: add, append, extend, delete
```

```
# Initial List
```

```
numbers = [10, 20, 30]
```

```
print("Initial List:", numbers)
```

```
# 1. append() - adds element at the end
```

```
numbers.append(40)
```

```
print("After append(40):", numbers)
```

2. insert() - adds element at a specific position

```
numbers.insert(1, 15)
```

```
print("After insert(1, 15):", numbers)
```

3. extend() - adds multiple elements

```
numbers.extend([50, 60])
```

```
print("After extend([50, 60]):", numbers)
```

4. remove() - deletes a specific element

```
numbers.remove(30)
```

```
print("After remove(30):", numbers)
```

5. pop() - deletes element at a specific index

```
numbers.pop(2)
```

```
print("After pop(2):", numbers)
```

6. del - deletes element using index

```
del numbers[0]
```

```
print("After del numbers[0]:", numbers)
```

Flowchart:



Output:

```
File Edit Format Run Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03)
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/hemal/OneDrive/Documents/SIMATS/AI/AI_...
Initial List: [10, 20, 30]
After append(40): [10, 20, 30, 40]
After insert(1, 15): [10, 15, 20, 30, 40]
After extend([50, 60]): [10, 15, 20, 30, 40, 50, 60]
After remove(30): [10, 15, 20, 40, 50, 60]
After pop(2): [10, 15, 40, 50, 60]
After del numbers[0]: [15, 40, 50, 60]
```

Result:

The program successfully demonstrates list methods such as adding, appending, extending, and deleting elements from a list.